ACE TELECOM · REPAIRDESK

Location Features Proposed App Wireframes

Geotagged job check-in and check-out, attendance by geofence, on-duty tracking, travel allowance, and live ETA for customers.

STATUS
Proposal — for review

PREPARED
30 September 2026

APPS
RepairDesk Android app (Technician, Staff, Customer) and Admin settings

MARKET
India — budget Android phones, patchy mobile data, DPDP Act 2023

These are low-fidelity wireframes built on the current app layout (blue app bar, Job Detail sections, Start Repair confirmation). Names, job numbers, distances and amounts are examples.

1. Overview

Five location features are proposed. Each one can be switched on independently by the Admin, and they are ordered so the simplest, most reliable features ship first.

#	Feature	What it does	Used by
1	Geotagged check-in / check-out	Records location and time when the technician taps Start Repair and Mark Complete. Proves the visit happened.	Technician, Staff
2	Attendance by geofence	Start day / End day allowed only near the shop. Replaces paper registers and biometric punching.	Technician, Staff, Admin
3	On-duty periodic tracking	Location every 5 minutes (configurable) between Start day and End day only. Live map for the dispatcher.	Staff
4	Distance and travel allowance	Calculates km travelled per day and per job, at the shop's per-km rate. One-tap claim and approval.	Technician, Staff, Admin
5	Customer live ETA	Customer sees the technician approaching after “On my way”, until check-in.	Customer

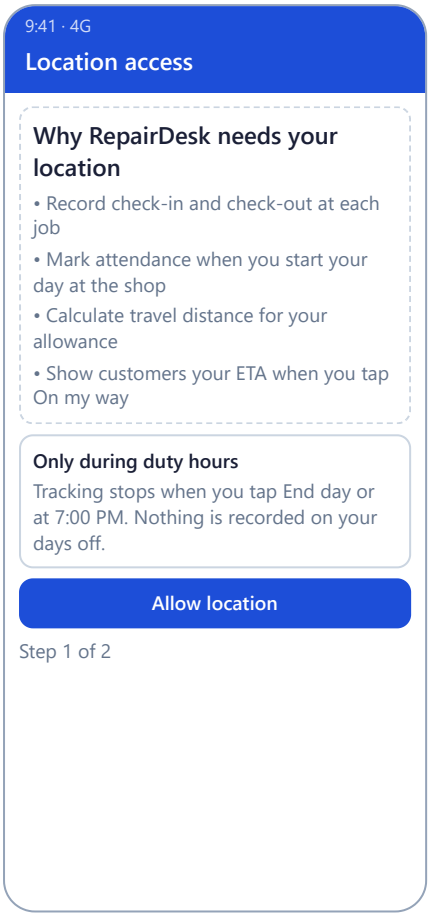
Screen	Role	Features	Page section
1A. Location explanation (first run)	Technician	All	2
1B. Android permission prompts	Technician	All	2
2. Home — Start day / End day	Technician	2, 3	2
3. Job — travel, check-in and check-out	Technician	1, 5	2
4. Travel & allowance	Technician	4	2
5. Live technician map	Staff	3	3
6. Proof of visit and attendance report	Staff / Admin	1, 2, 4	3
7. Track your technician	Customer	5	4
8. Location settings	Admin	All	5

Nothing changes for how technicians work a job today. Check-in and check-out ride on the existing **Start Repair** and **Mark Complete** buttons, so there are no new steps to learn for feature 1.

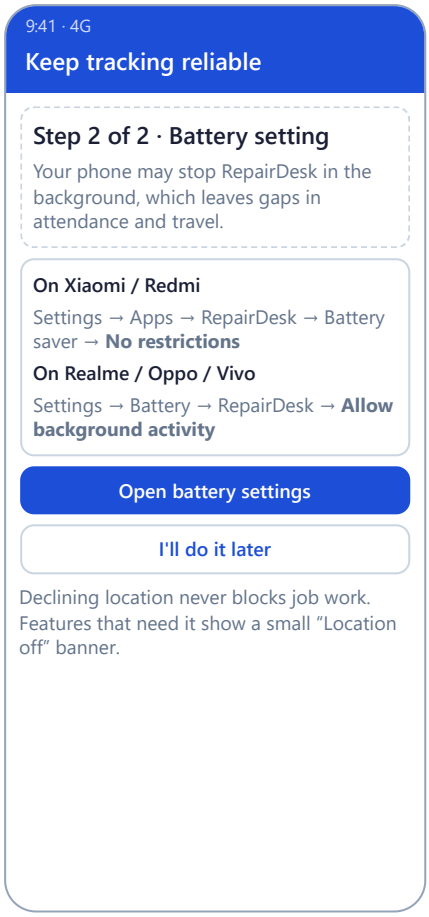
2. Technician app

Screen 1A — First run: RepairDesk explains location use

All features



Shown once after first sign-in, before Android's own prompt.

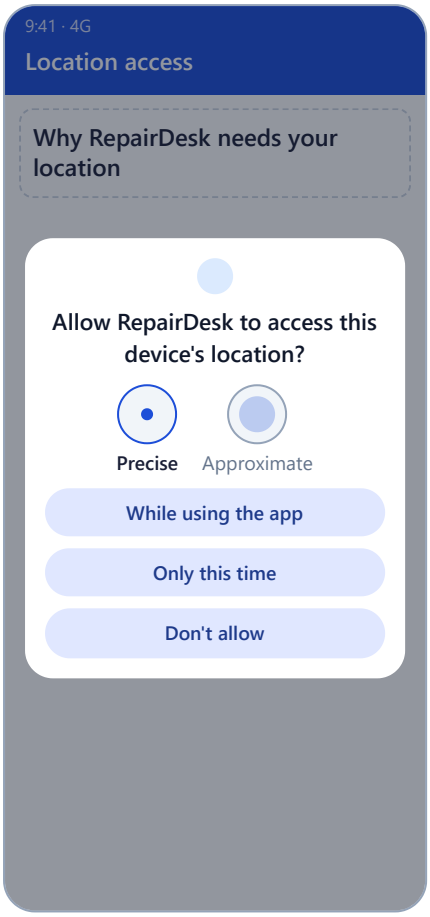


Brand-specific help, deep-linking to the right settings page where possible.

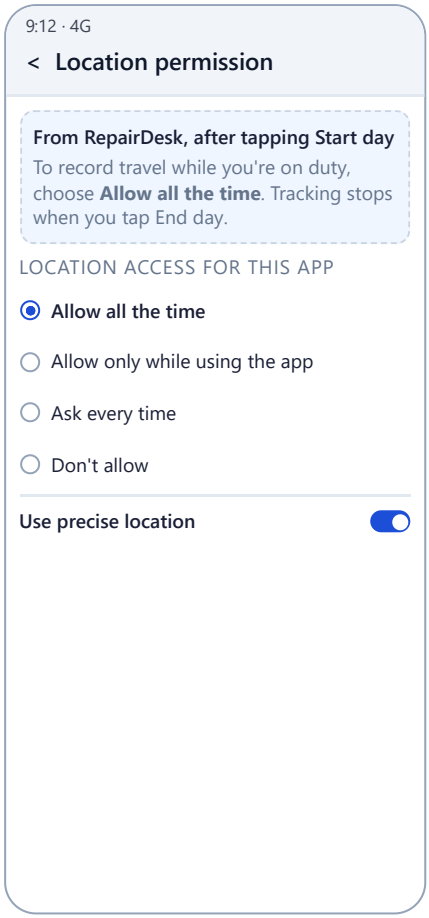
1. Every use of location is explained before the Android prompt, as the Digital Personal Data Protection Act, 2023 expects a clear notice and purpose.
2. "While using the app" is requested first. Background location ("Allow all the time") is only needed for tracking and distance, so it is requested when the technician first taps Start day.
3. The battery step matters most on Xiaomi, Realme, Vivo and Oppo phones, which aggressively stop background apps.
4. Android never grants location at install time. Android's own prompts (Screen 1B) appear when the technician taps **Allow location**.

Screen 1B — Android's permission prompts

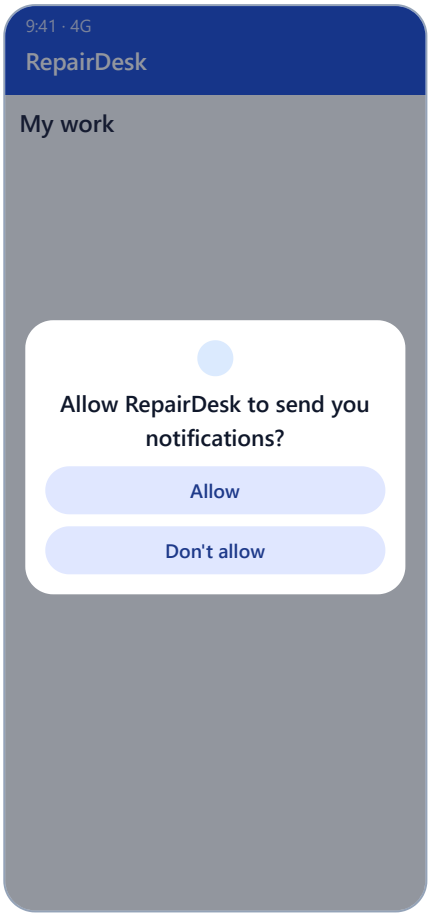
All features · Android 12 and newer



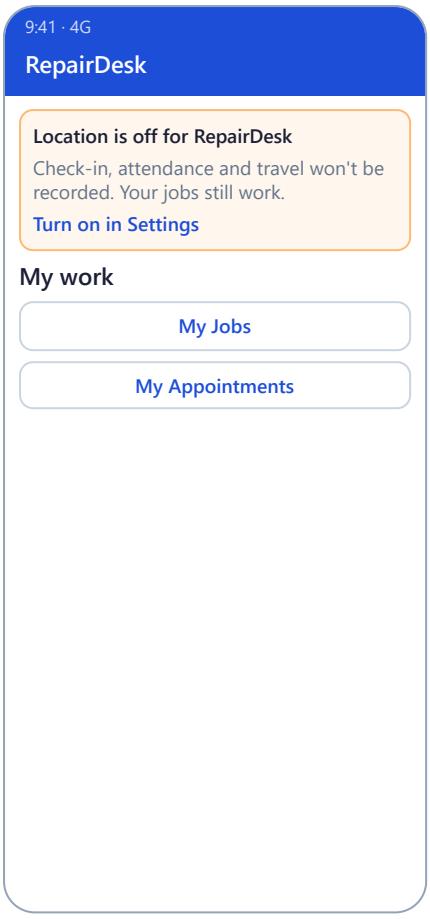
Android's first prompt. There is no "Allow all the time" here on Android 11+.



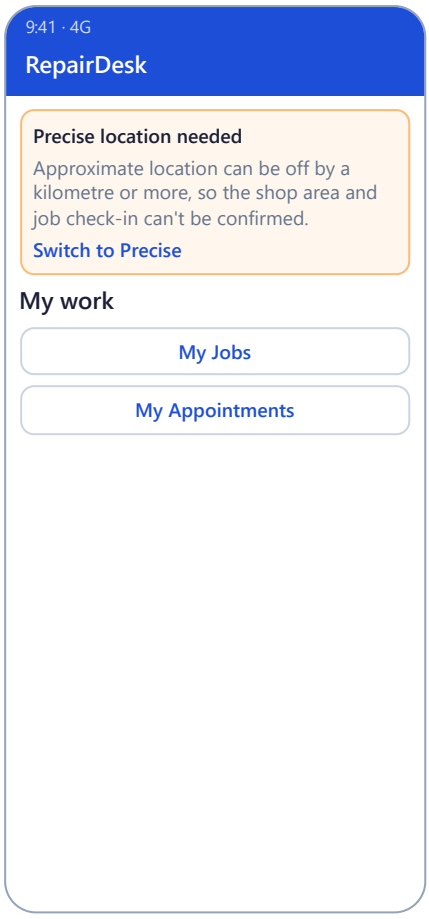
Background location is chosen in Android Settings; RepairDesk opens this page for the technician.



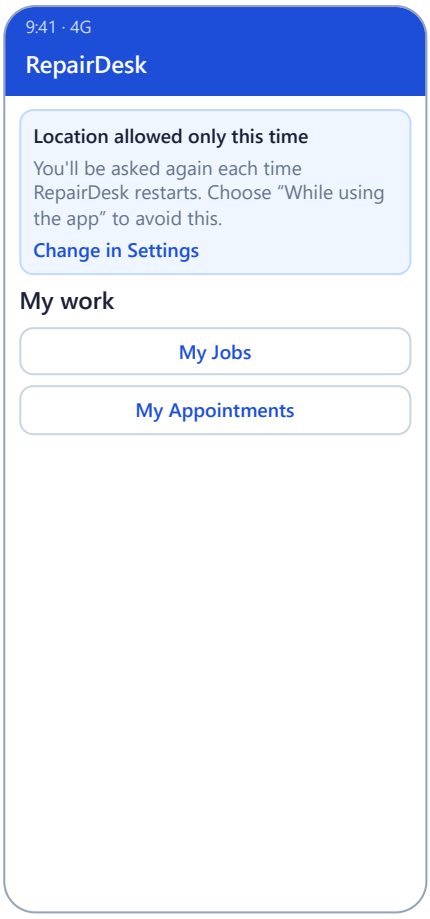
Android 13+. Needed for job alerts and the "sharing location while on duty" notification.



After "Don't allow". Declined twice, Android stops prompting, so the banner opens Settings.



After choosing Approximate in the first prompt.



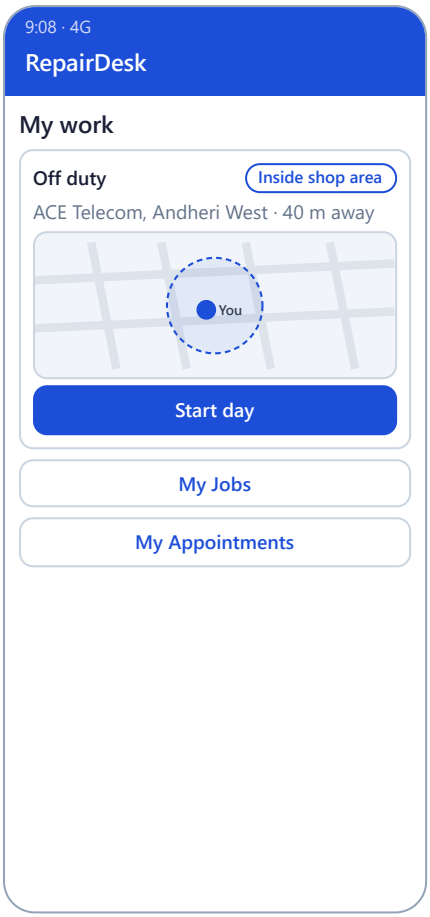
After "Only this time".

Technician chooses	What works in RepairDesk
While using the app + Precise	Check-in / check-out, Start day geofence, On my way ETA
+ Allow all the time (Settings)	Also on-duty tracking and distance for allowance
Approximate	Too inaccurate for geofence and check-in — app asks to switch to Precise
Only this time	Works until the app restarts, then the prompt returns
Don't allow	Jobs still work; location features show a "Location off" banner

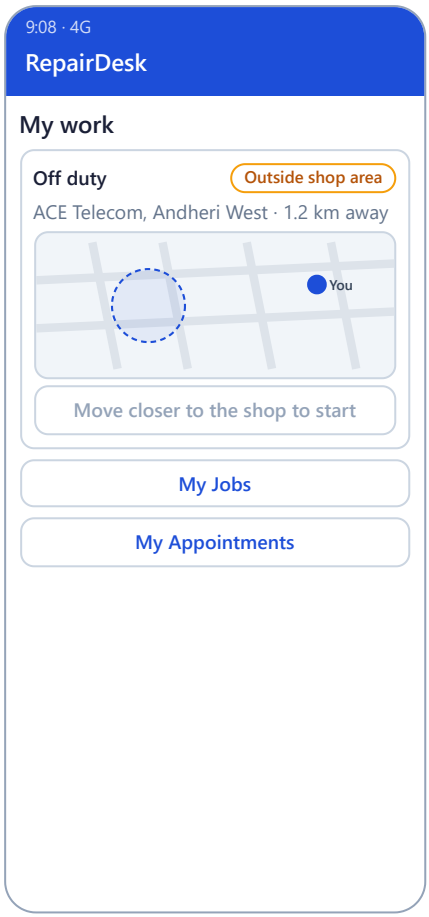
1. Older phones differ: Android 10 offers Allow all the time / Allow only while using the app / Deny in one prompt; Android 6–9 show only Allow / Deny.
2. On company-owned phones, location can be pre-granted with `adb install -g` or a device-management tool (for example Microsoft Intune), so no prompt appears.
3. If RepairDesk is published on the Play Store, Google requires the Screen 1A explanation plus a background-location declaration and demo video.

Screen 2 — Home: start and end the day

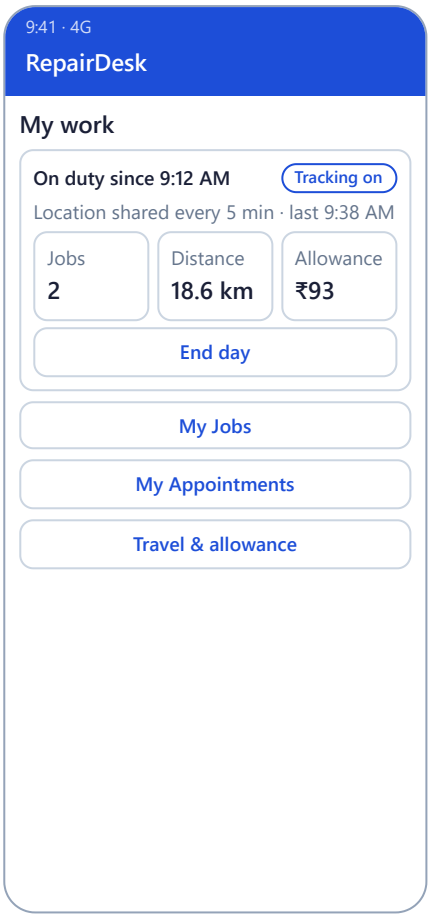
Attendance by geofence · On-duty tracking



Before duty. Start day works only inside the shop geofence.



Outside the area the button is disabled (unless the Admin allows field starts).

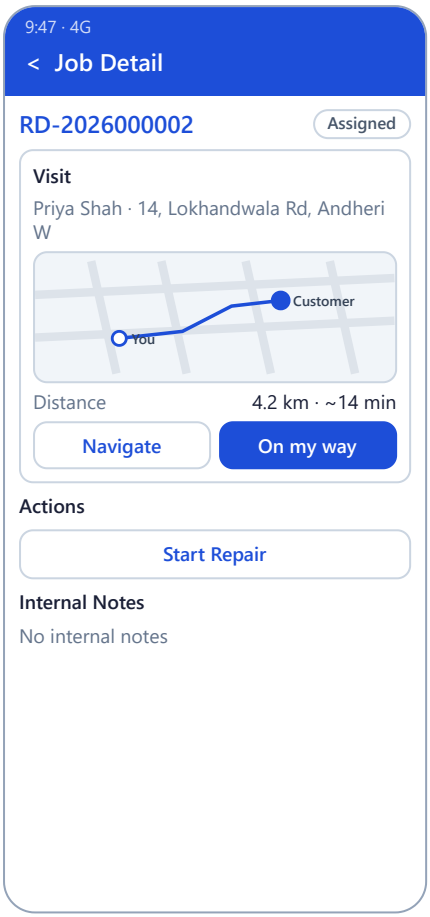


On duty. A persistent Android notification shows tracking is on.

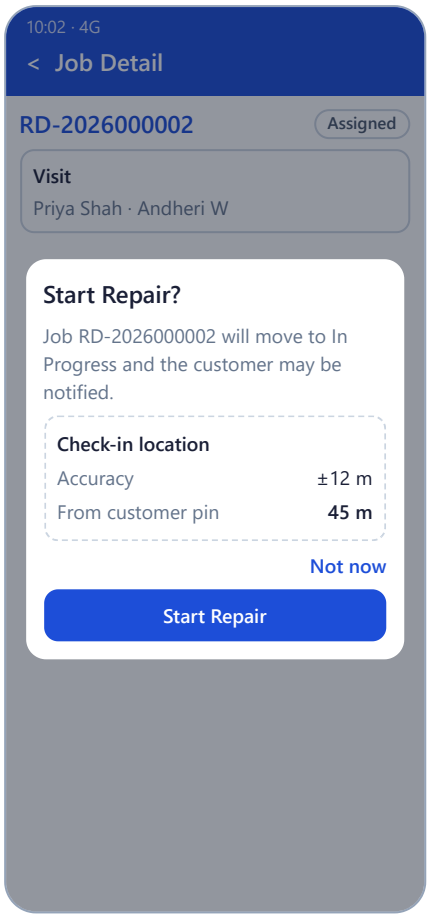
1. Start day records attendance time and location; allowed only within the shop radius set by the Admin (for example 150 m).
2. Tracking runs only between Start day and End day. If the technician forgets End day, it stops automatically at the working-hours end time.
3. Android requires an ongoing notification (“RepairDesk is sharing your location while on duty”) for background tracking — it also reassures the technician.

Screen 3 — Job: travel, check-in and check-out

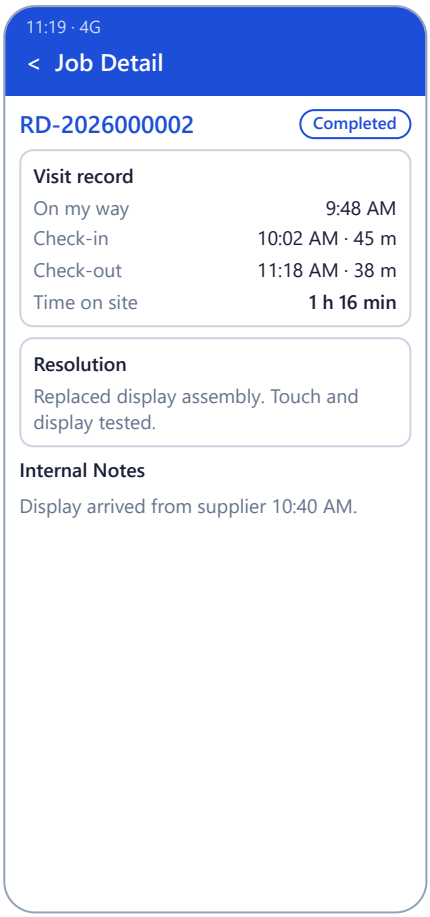
Geotagged check-in / check-out · Live ETA



New Visit section. On my way starts ETA sharing with the customer.



The existing Start Repair confirmation also captures the check-in.

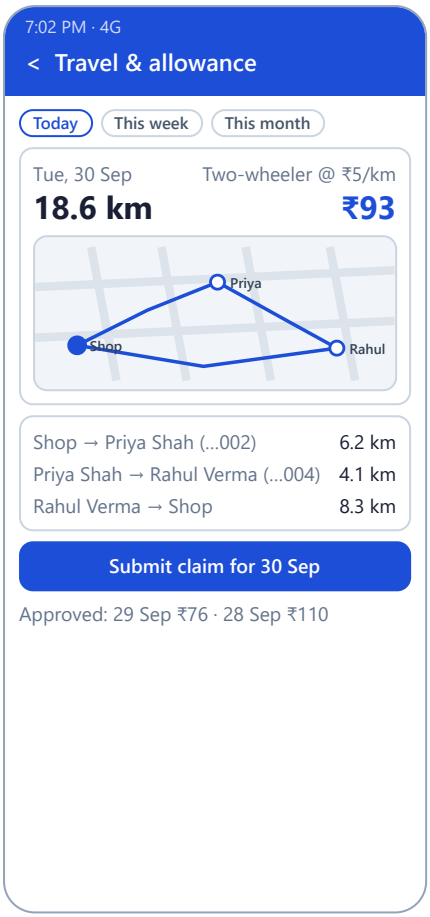


After Mark Complete, the Visit record is the proof of visit.

1. Check-in is recorded on confirming Start Repair; check-out on Mark Complete. Distances are measured from the customer's saved pin.
2. The customer pin is saved on the first visit, because Indian addresses are often landmark-based.
3. With weak GPS (basements, older buildings) the app takes the best fix within 20 seconds and marks it “Low accuracy” instead of blocking the job.
4. On my way is optional; sharing stops automatically at check-in. Bench jobs at the shop skip the Visit section.

Screen 4 — Travel & allowance

Distance travelled · Conveyance allowance



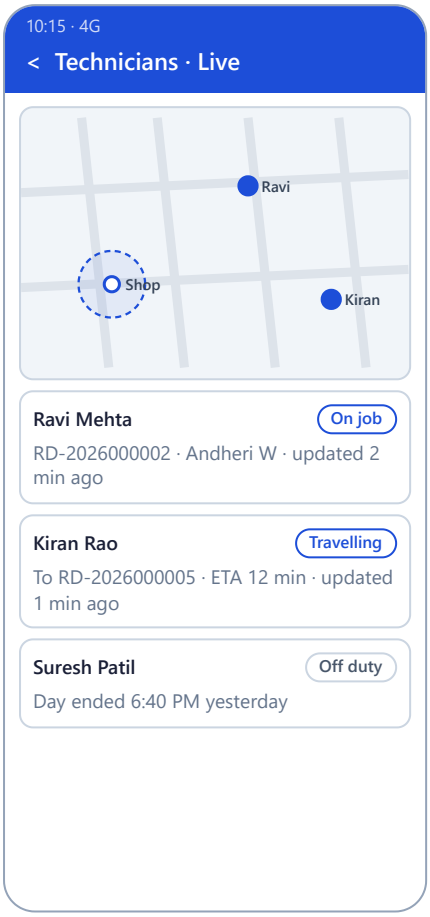
Opened from home. Uses on-duty location points only.

1. Distance is road distance between on-duty points, with GPS jumps filtered out (speeds over 120 km/h, or jitter while parked).
2. Trips are split at job check-ins, so each leg is tied to a job reference — this is what makes the claim quick to approve.
3. Rate and vehicle type come from Admin settings. Technicians cannot edit the distance but can add a note if a reading looks wrong.
4. One claim per day. Staff or Admin approve it from the Attendance & travel report (Screen 6).

3. Staff / dispatcher

Screen 5 — Live technician map

On-duty tracking

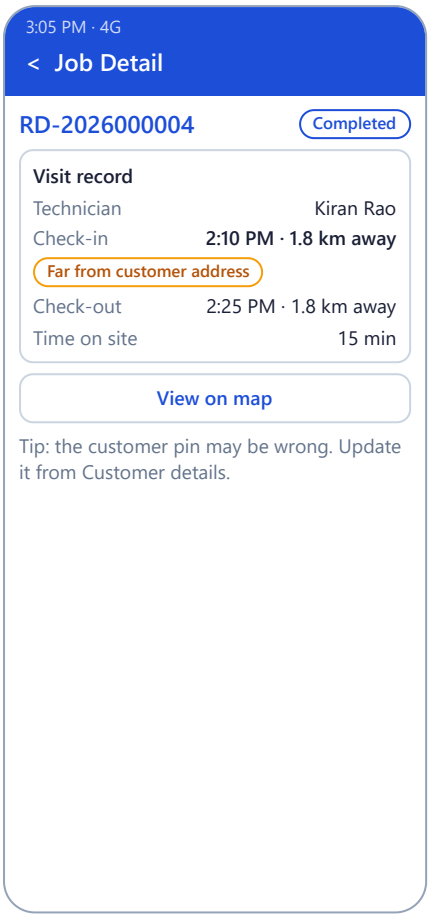


New quick action on the Staff/Admin dashboard.

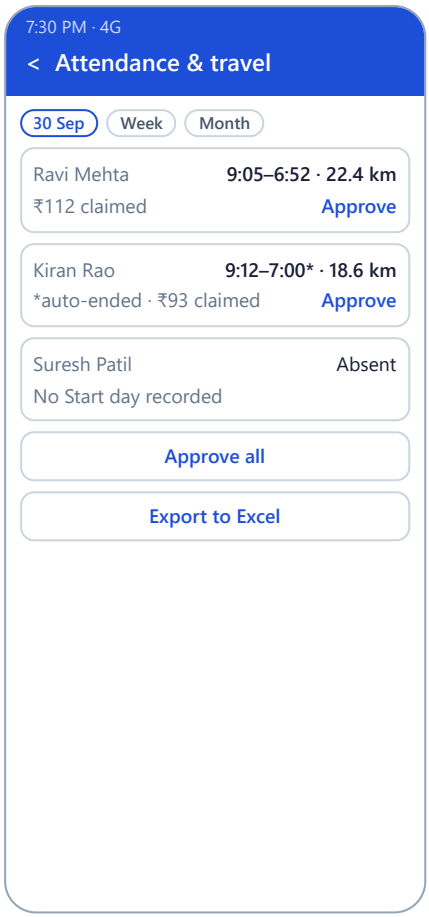
1. Only on-duty technicians appear on the map. Off-duty technicians show a status, never a location.
2. “Updated x min ago” turns into a warning after 15 minutes without a ping — usually the phone stopped the app or has no network.
3. Tapping a technician opens today's timeline: Start day, each check-in and check-out, and the current job.
4. When assigning a job, the Assign dialog can sort technicians by distance to the job address.

Screen 6 — Proof of visit and attendance report

Check-in / check-out · Attendance · Allowance approval



Staff see the Visit record, flagged when check-in was far from the address.



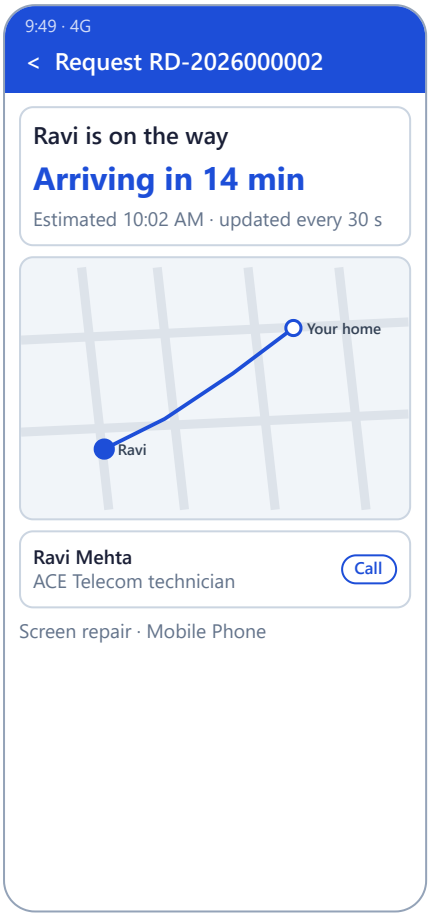
Daily report for Staff/Admin; export for payroll.

1. The “far from address” flag (default over 500 m) is informational. A wrong customer pin is the most common cause.
2. The report replaces the paper register: first Start day, last End day, and whether each was inside the shop area.
3. Claims are approved one by one or in bulk; approved amounts appear in the monthly export.

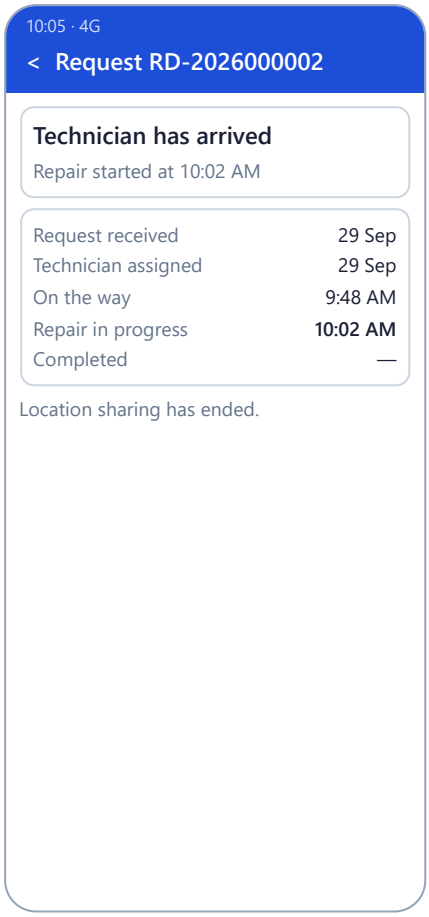
4. Customer app

Screen 7 — Track your technician

Live ETA



Opens from the "Ravi is on the way" notification or SMS link.



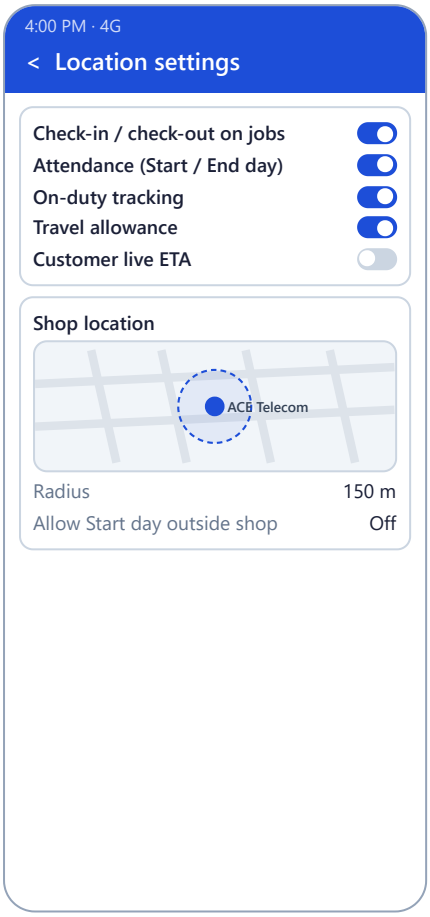
After check-in the map is replaced by the status timeline.

1. The customer sees the technician's location only between On my way and check-in — never before or after.
2. The SMS link opens a lightweight web page, so customers without the app can track too.
3. The Call button can later switch to masked calling so the technician's number is not shared.

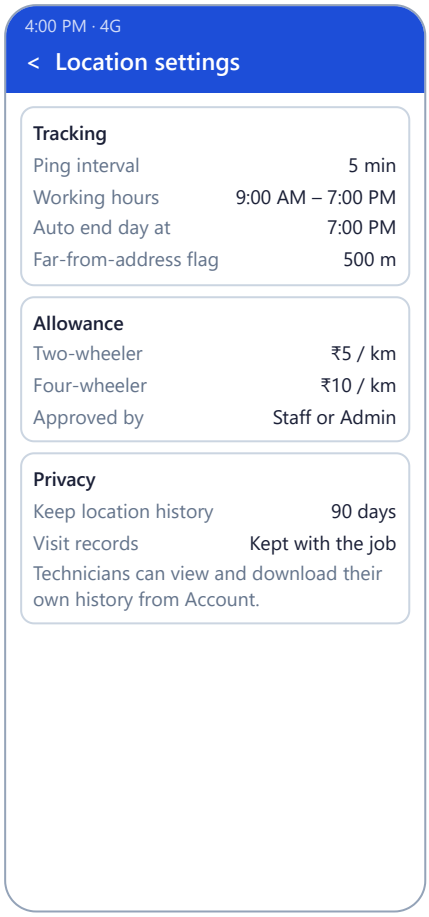
5. Admin settings

Screen 8 — Location settings

Controls for all five features



Each feature is switched on independently.



Tracking, allowance and retention rules.

1. Every rate and threshold is a setting, so each shop can match its existing allowance policy.
2. A 2-minute interval improves the live map but uses more battery; 5 minutes is a good default on budget phones.
3. Raw location points are deleted after the retention period. Visit records stay with the job as proof of service.
4. Branches can be added later, each with its own geofence.

6. Rollout, privacy and practicalities

Suggested rollout

Phase	Feature	Screens	Background location
1	Geotagged check-in / check-out	3, 6, 8	No — one fix when a button is tapped
2	Attendance by geofence	1, 2, 6, 8	No
3	On-duty tracking	2, 5, 8	Yes
4	Distance and travel allowance	4, 6, 8	Yes
5	Customer live ETA	3, 7, 8	Yes, only while travelling to a job

Phases 1 and 2 need only a single location reading when the technician taps a button, so they work reliably on budget phones and raise few privacy concerns. Phases 3–5 need background tracking, a foreground notification, and battery-setting guidance, so they follow once the basics are proven.

What is recorded, and for how long

Data	When	Who sees it	Kept
Check-in / check-out point	Start Repair, Mark Complete	Staff, Admin, the technician	With the job
Start day / End day point	Tapping Start day / End day	Staff, Admin, the technician	Retention period
On-duty location points	Every 5 min while on duty	Staff, Admin (live map, distance)	Retention period (90 days default)
Live position for customer	On my way until check-in	That job's customer only	Not stored beyond the trip

India-specific practicalities

Topic	Approach
Budget Android phones	Foreground service with a visible notification, plus brand-specific battery-setting help (Screen 1).
Patchy mobile data	Points are stored on the phone and uploaded when the network returns; job actions never wait for GPS.
Data and battery use	About 1–5 MB of data per month at a 5-minute interval; a few percent of battery per day.
Landmark addresses	Save the customer's pin on the first visit and reuse it.
Maps cost	Google Maps by default; Mappls (MapmyIndia) or Ola Maps are cheaper alternatives if usage grows.
DPDP Act, 2023	Clear notice and purpose, duty hours only, a retention limit, and technicians can see their own data.

Decisions needed

Question	Proposed default
Shop geofence radius	150 m
Can technicians start the day from home or the field?	Off; Admin can enable per shop
Tracking interval	5 minutes
Allowance rates	₹5 / km two-wheeler, ₹10 / km four-wheeler
Who approves allowance claims	Staff or Admin
Location history retention	90 days
Customer live ETA at launch	Off until phases 1–4 are stable